Neel Shah

Applied Machine Learning · Model Validation · ML Engineering

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SUMMARY

I build machine learning systems and the validation tooling that decides whether they ship. My project, CourtVision, covers four sports (NBA, MLB, soccer, tennis): per-stat **XGBoost** models with **isotonic calibration**, a player-prop stack scored on a 20,354 player-game chronological holdout, in-game models conditioned on live game state, and a broadcast-video tracking pipeline. Every model is scored **leak-free and walk-forward**; the same harness caught and retracted six of my own inflated results, and 0 of 60 candidate signal classes passed it.

SELECTED PROJECT: COURTVISION

Calibrated multi-sport forecasting engine, with the validation harness and paper-only execution controls as first-class components.
Evidence: github.com/neeljshah/court-vision/blob/master/docs/QUANT_BRIEF.md · github.com/neeljshah/court-vision/blob/master/EVIDENCE.md

Models and results

- **Modeling and data:** per-stat **XGBoost** with **isotonic calibration**; four de-vig methods from scratch incl. Shin (1992); a 151-file intelligence layer incl. a 291,625-pair player-vs-player matchup matrix; AI agents propose signals and the gate rejects them.
- **Player props (NBA):** chronological holdout of 20,354 player-games, **MAE** PTS 4.83, REB 1.92, AST 1.39, FG3M 0.89; reported as accuracy only, with no market comparison.
- **In-game:** conditioning on realized game state is sharper than a static pregame prior (NBA Brier 0.209 to 0.159, MLB 0.241 to 0.126, real-corpus out-of-sample). Where a live market exists (NBA end of Q1, Kalshi MLB and soccer ticks), the market is sharper than the model, and the verdicts are published.
- **Possession Monte Carlo:** a player-level basketball simulator where teammates compete for a shared scoring pie sampled from real stint minutes, so teammate correlation emerges from the mechanics instead of a hand-tuned matrix.

Validation

- **Harness:** expanding-window **walk-forward** with a per-fold date assertion, **truncation-invariance** leak tests, acceptance only when a change beats baseline on **two independent corpora**, and a published minimum detectable effect for every market-relative row.
- **Overfit caught:** found a leaky grid search in my own prop pipeline (CV R2 about **0.79** vs leak-free holdout about **0.06**) and hard-coded the corrective regularization.
- **Published results:** pregame probabilities match or trail the devigged close and the in-game model is behind the live market, with n and 95% intervals; six withdrawn figures published with reasons at /evidence/retractions/; 16 findings on one template with limitations and verdict history.
- **Signals:** 0 of 60 candidate signal classes shipped; 513 verdicts were REJECT or DEFER. Candidate-specific gates included clustered DM, confidence intervals, null-shuffle tests, and **Benjamini-Hochberg** control.

Engineering

- **Private engine: FastAPI** with ~99 endpoints across 12 routers, 9 long-running daemons under a watchdog, a 430-module codebase, **~7,400-test pytest suite**, CI/CD, 5 Docker images; an MCP server (interface documented publicly) answers queries from AI agents with cited artifacts.
- **Computer vision:** broadcast video to court coordinates on one consumer GPU: **YOLOv8** detection, SIFT homography, Kalman+Hungarian tracking, OSNet re-ID reimplemented in **PyTorch**; output quality measured and documented, and tracking paused because its features carry zero importance in the production models.
- **Evidence site:** one unified, CI-gated site at https://neeljshah.github.io/court-vision/, built by one person directing AI coding agents; performance-budget and accessibility (axe) gates are hard CI steps with zero serious or critical violations; a PAPER ONLY desk record with receipts.

PROFESSIONAL EXPERIENCE

Data Analyst Intern, Sunsolor · *Tempe, AZ · 2025*
Built live Tableau / Power BI suites; cut manual reporting latency ~30%; automated daily market-trend pipelines in Python; data-quality audits across 5+ datasets; optimized SQL on 1M+-row datasets.

Data Annotation Specialist, DataAnnotation.Tech · *Remote · 2024–2025*
Labeled 10,000+ structured samples for production ML training & LLM fine-tuning; authored corner-case documentation cutting pipeline error variance ~20%; refined labeling schemas with data scientists.

Business Intelligence Analyst, Fortrex Securities · *Madison, WI · 2023–2024*
Built secure executive reporting interfaces (99.9% uptime); refined backend payment APIs & reporting pipelines for high-throughput billing; engineered real-time anomaly-detection alerting.

EDUCATION & CERTIFICATIONS

University of Iowa: three years of data-science coursework. Coursework: math stats, probability theory, machine learning, financial analytics.
Certifications: Certified Data Scientist (USDSI) · ML Specialist & Data Viz (Kaggle) · CBIAP (IABAC).

SKILLS

ML/CV: PyTorch, OpenCV, YOLOv8/Ultralytics, ONNX, TensorRT, NumPy, Pandas, scikit-learn, XGBoost, LightGBM, statsmodels
Languages: Python (expert), SQL, JavaScript/TypeScript (working)
Methods: walk-forward / leak-free validation, calibration (isotonic), de-vigging (Shin), fractional-Kelly, Monte-Carlo simulation, shadow-logging, multi-agent orchestration
Backend/Infra: FastAPI, WebSocket/SSE, PostgreSQL, SQLite, Docker, GitHub Actions, GCP, Tableau, Power BI, dashboards